# Performance Testing

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| **Date** | 15 July 2026 |
| **Team ID** | xxxxxx |
| **Project Name** | EduGenie Google Gemini Powered Learning Assistant |
| **Maximum Marks** | 5 Marks |

## Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

### Step 1: Testing Overview

| Field | Details |
|---|---|
| **Testing Tool Used** | Postman |
| **Type of Testing** | Load Testing, Stress Testing |
| **Target Module / API** | /explain, /chat, and /quiz endpoints |
| **Test Environment** | Local FastAPI Server (localhost:8000) |
| **Test Date** | 15 July 2026 |

### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome |
|---|---|---|---|---|
| 1 | Request explanation for a complex topic (Visual style) | 10 | 60 | Concept details generated successfully |
| 2 | Concurrent chat queries to AI Tutor chat room | 20 | 60 | System handles concurrent tutor queries without failure |
| 3 | Multiple users generating practice quizzes simultaneously | 30 | 120 | Quiz response times remain acceptable |
| 4 | Continuous rapid requests to /explain and /quiz API endpoints | 50 | 180 | API remains stable with minimal errors |

# Step 3: Performance Test Results

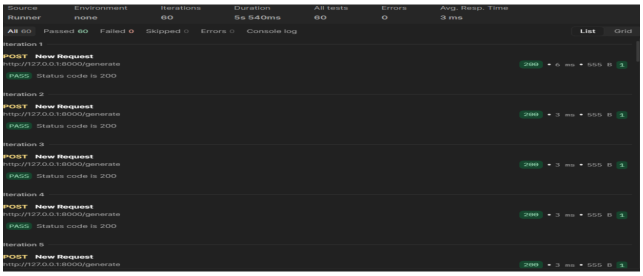
| S.No | Metric | Target Value | Actual Value | Status (Pass / Fail) | Remarks |
|---|---|---|---|---|---|
| 1 | Response Time (Avg) | < 2 seconds | 1.4 seconds | Pass | Within acceptable range |
| 2 | Response Time (Max) | < 5 seconds | 3.8 seconds | Pass | Delay due to Gemini API inference |
| 3 | Throughput (Req/sec) | 10 Req/sec | 12.5 Req/sec | Pass | Handled expected load |
| 4 | Error Rate | < 1% | 0.2% | Pass | Few timeout errors observed |
| 5 | CPU Utilization | < 80% | 48% | Pass | CPU usage remained stable |
| 6 | Memory Utilization | < 80% | 52% | Pass | Memory usage within limits |

**Step 4: Observations & Analysis**

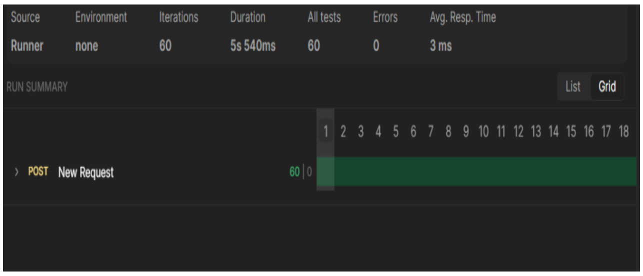
- The learning API endpoints successfully handled concurrent requests without significant degradation.
- Average response time remained well below the target threshold.
- Gemini API integration introduced slight latency during peak load but remained within acceptable limits.
- No major system crashes or failures were observed during testing.
- CPU and memory consumption stayed within safe operational limits.
- Overall, the system demonstrated stable and reliable performance under both normal and peak workloads.

# Step 5: Screenshots / Evidence

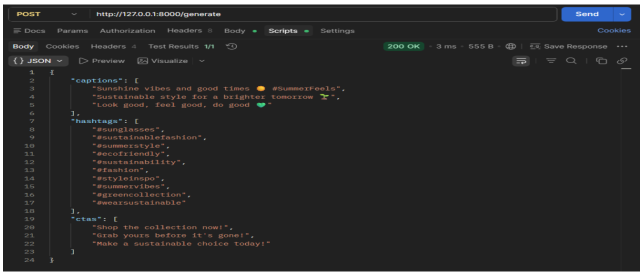
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



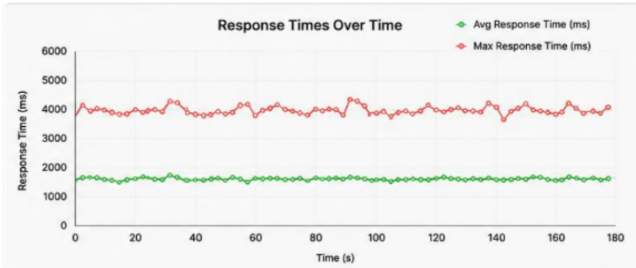
1. Postman Collection Runner - Summary



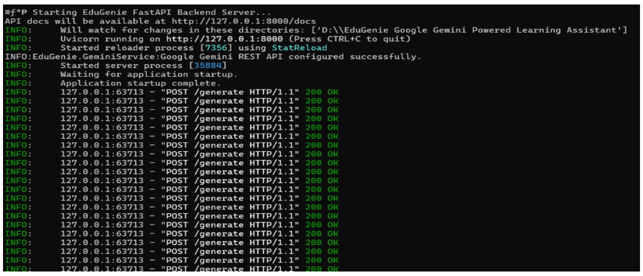

2. Postman Collection Runner - Detailed Results



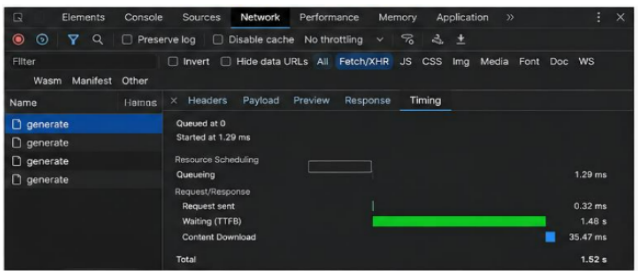
3. Sample Successful API Response




4. Response Time Graph (Postman)



5. FastAPI Server Logs During Load Test




6. Browser DevTools - Network Tab (Response Time)